

DOES CLUSTER LOADING ENHANCE LOWER BODY POWER DEVELOPMENT IN PRESEASON PREPARATION OF ELITE RUGBY UNION PLAYERS?

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ABSTRACT

Hansen, KT, Cronin, JB, Pickering, SL, Newton, MJ. Does cluster loading enhance lower body power development in preseason preparation of elite rugby union players? *J Strength Cond Res* 25(8): 2118–2126, 2011—The purpose of this study was to ascertain whether cluster training led to improved power training adaptations in the preseason preparation of elite level rugby union players. Eighteen highly trained athletes were divided into 2 training groups, a traditional training (TT, $N = 9$) group and a cluster training (CT, $N = 9$) group before undertaking 8 weeks of lower body resistance training. Force-velocity-power profiling in the jump squat movement was undertaken, and maximum strength was assessed in the back squat before and after the training intervention. Two-way analysis of variance and magnitude-based inferences were used to assess changes in maximum strength and force, velocity, and power values pretraining to posttraining. Both TT and CT groups significantly ($p < 0.05$) increased maximum strength posttraining. There was a possibly negative effect for the CT group on maximum strength when compared with that for the TT group (pretraining to posttraining change = 14.6 ± 18.0 and $18.3 \pm 10.1\%$, respectively). There were no significant differences pretraining to posttraining for any jump squat force, velocity, or power measures. However, magnitude-based inferences showed that there was a likely positive effect of CT when compared with that of TT for peak power (pretraining to posttraining change = 7.5 ± 7.0 and $1.0 \pm 6.2\%$, respectively) and peak velocity at 40 kg (pretraining to posttraining change = 4.7 ± 6.1 and $0.0 \pm 5.0\%$, respectively) and for peak velocity at body weight (pretraining to posttraining change = 3.8 ± 3.4 and $0.5 \pm 3.8\%$, respectively). Although both a traditional and cluster training loading pattern improved

lower body maximum strength in a highly trained population, the traditional training structure resulted in greater maximum strength adaptation. There was some evidence to support the possible benefit of cluster type loading in training prescription for lower body power development.

KEY WORDS power, jump squat, cluster sets

INTRODUCTION

Strength and power are physical attributes that have been shown to be crucial to high-level performance in collision sports such as rugby league, rugby union, and American Football (2,5). The development of strength and power is therefore an important component of training programs for the preparation and development of elite level athletes in these sports. Given the complex nature of resistance training prescription for collision sports, training interventions require careful consideration to ensure that training outcomes are achieved. Cluster loading, sometimes termed interrepetition rest training, describes a training system whereby the rest periods are manipulated, breaking sets into small clusters of repetitions (19,29,30). It has been previously suggested that these training structures may be well suited to the development of lower body explosive performance (19) and thus may be appropriate for use in collision sport athletes.

There are a number of factors inherent in the preparation of high-level rugby union and rugby league players, which make training prescription for power development complex for the practitioner. First, these sports are characterized by long in-season periods (typically 25–35 weeks) with relatively short preseason preparations (typically 8–12 weeks) (3,18,23). Thus, the preseason preparatory period wherein resistance training frequency and volume can be increased represents a relatively short time frame for the development of lower body power, particularly in highly trained athletes. Additionally, in many collision sports, there is the added complexity of the considerable demands of other training components such as metabolic conditioning, speed and skill development, and team organization (3,11). This additional training imposes

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many different physiological demands on the athlete, which can adversely affect power development, (23,24). Therefore, the investigation of strength and power interventions applied in this context is crucial to improving the understanding of athletic development in these and other similar sports. Although the design of strength and power programs in rugby league and rugby union has been the subject of considerable research (1,3,8), to the authors' knowledge, no studies have investigated the use of cluster loading in collision sports.

It has been postulated that breaking sets into small "clusters" of repetitions may improve the kinematic and kinetic (force, power, velocity) profile of a training set. This in turn may lead to improved training outcomes, particularly in the training of ballistic performance (19). To this end, investigations into the acute effect of cluster structures have suggested an improved set velocity and power profile during both lower body and upper body movements (14,20,30). These improvements have been attributed to the ability of the short rest periods between clusters to allow metabolic recovery resulting in improved kinematics and kinetics in the latter repetitions of the set when compared with traditional loading paradigms. This improved set profile may be beneficial because there is some evidence that adaptation to ballistic performance may be principally mediated by neural mechanisms, with intramuscular (21,32) and intermuscular (33) neural adaptations contributing to performance improvements after high velocity training.

However, there is very little research tracking training outcomes after the implementation of cluster loading programs. Studies have suggested that various cluster loading configurations in untrained subjects confers no beneficial effect in terms of maximum strength adaptation when compared with traditional training structures (10,35). In elite junior basketball players, Lawton et al. (29) compared upper body strength and power adaptations in the bench press movement between a cluster intervention and a traditional training structure over a 6-week training period. This research showed that a traditional structure resulted in significantly greater gains in maximal strength (9.7%) compared with the cluster structure (4.9%), but there were no significant differences in power adaptation between interventions. However, there remains no published research with cluster set structures applied to lower body power training when ballistic movements are included.

The purpose of this study was to investigate whether cluster arrangements led to improved training adaptations when compared with a traditional set structure during the pre-season preparation of highly trained elite level rugby union players. Despite suggestions that cluster loading is well suited to the development of mechanical power, to date, there is limited research investigating training outcomes with these set configurations applied alongside loading parameters commonly used in the training of mechanical power in athletes. This study addressed this gap in the research using

highly trained rugby union players for whom lower body power is a key physical attribute.

METHODS

Experimental Approach to the Problem

To compare traditional and cluster loading for the development of strength and power, 18 elite rugby union players undertook 8 weeks of resistance training using the squat and clean movement patterns. The players were randomly allocated to 1 of 2 groups, a traditional training (TT) group and a cluster training (CT) group. Training was undertaken during the preseason training phase, which represents the time of the year when their greatest resistance training volume is typically undertaken. To ascertain the effect of the training interventions on lower body strength and power performance, preceding and after the training intervention players undertook force, velocity, and power profiling of the jump squat at a variety of light to moderate external loads and maximum strength was assessed in the back squat movement. Training outcomes were evaluated using effect statistics and percentage change in maximum strength, force, velocity, and power. Differences in training outcomes between groups were assessed using 2-way analysis of variance (ANOVA) and 90% confidence limits (CLs) from which a qualitative inference of the effect of the cluster intervention was derived.

Subjects

Eighteen highly trained elite male rugby union players undertaking preseason training before the start of their competitive season agreed to participate in this study. This represented the total number of available subjects who fulfilled the study inclusion criteria. These criteria were as follows: (a) the athlete was scheduled to be present for the entire training block; (b) the athletes' individual training goals agreed by conditioning and coaching staff and the athlete were congruent with the training prescription for the study; and (c) the study prescription was deemed appropriate for the athlete considering musculoskeletal screening results and injury history. The average age and height were 26.8 ± 4.5 years and 1.89 ± 0.1 m, respectively, and the average body mass was 103.5 ± 8.6 and 104.3 ± 8.5 kg pretraining and posttraining, respectively. All the subjects had the procedures, benefits, and risks of participation explained to them and provided informed consent. All the procedures were approved by the Edith Cowan University Human Research Ethics Committee.

Procedures

Before starting the training intervention and at the completion of the training intervention, the subjects undertook assessment of jump squat force, velocity, and power across a spectrum of loads and back squat maximum strength testing. Jump squat testing and strength testing took place on separate days at least 48 hours apart with jump squat assessment preceding maximum strength assessment. All the athletes had 3 weeks of active rest at the completion of the previous

competitive season followed by 3 weeks of prescribed self-directed preparatory strength and conditioning before the study commenced.

Power Testing Procedures. After a standardized warm-up, each subject performed 3 jump squats at body weight and 3 external loads, 20, 40, and 60 kg (in a randomized order) using a technique identical to that described by Hori et al. (26). Absolute loads were chosen for analysis because of primary interest were the changes in the athletes' ability to apply power to an absolute load irrespective of changes in body weight and maximum strength during the course of the training period. A similar spectrum of loads has previously been used in an investigation of power training in novice subjects (12) and in the assessment of lower body mechanical power of collision sport athletes (3). The jump technique involved the subjects standing at a self-selected foot width with an Olympic bar placed on their upper trapezius immediately below C7. The subjects then performed a countermovement to a self-selected depth and immediately preformed a maximal jump. The subjects were instructed to keep the depth of countermovement consistent between jumps and "jump for maximum height" on each repetition. All the subjects were familiar with the jump squat movement because they had previously performed it as part of both the training and testing programs.

All jumps were performed on a portable force plate (Accupower, AMTI, Watertown, MA, USA). Ground reaction force (GRF) data were sampled at 500 Hz via an analog to digital converter (National Instruments, Austin, TX, USA) and collected by means of a laptop computer using custom-built data acquisition software (Labview 8.2, National Instruments). Data were then transferred to a customized data analysis program for calculation of the kinematic and kinetic variables of interest (LabView 8.2, National Instruments).

Strength Testing Procedures. Maximum strength was assessed through predicting back squat 1 repetition maximum (1RM) from a 2 to 6 RM lift. Methods were similar to those previously outlined for the assessment of squat maximum strength in professional rugby union players (1). This involved each athlete performing 3 sets of 2–6 repetitions at incrementally increasing loads before 1 set was performed to failure in the 2–6 repetition range. Each repetition was performed to a visually assessed knee angle of 90°. One RM was then predicted using a documented equation (9). This calculation method has been shown to have a very high correlation ($r = 0.97$) to actual back squat 1RM (31). Our data show the methodology is a reliable means of assessing back squat 1RM in the study population (intraclass correlation coefficient [ICC] = 0.90, coefficient of variation [CV] = 5.9%).

Jump Squat Data Analysis. Power applied to and the velocity of the center of mass of the system were calculated from GRF data using the impulse momentum (forwards dynamics) approach outlined previously in the literature (15). Because

the initial velocity of the system was zero, at each time point throughout the jump, vertical GRF was divided by the mass of the system to calculate the acceleration of the system. Acceleration due to gravity was then subtracted such that only the acceleration generated by the subject was multiplied by time data to calculate instantaneous velocity of the system's center of mass. The resultant velocity data were then multiplied by the original GRF data to calculate power. From the resultant force-time, velocity-time, and power-time curves, the following 3 variables were calculated for each repetition: (a) Peak Force (PF) refers to the highest point on the force-time curve calculated from GRF data (between-day reliability, ICC = 0.96, CV = 2.3%). (b) Peak power (PP) refers to the highest point on the power-time curve calculated from GRF data (between-day reliability, ICC = 0.94, CV = 4.6%). (c) Peak velocity (PV) refers to the highest point on the velocity-time curve calculated from GRF data (between-day reliability, ICC = 0.93, CV = 3.4%).

Training Intervention. The subjects were randomly allocated to either a TT group ($N = 9$) or a CT group ($N = 9$), which used a cluster loading pattern. Descriptive statistics for each group can be observed from Table 1. There were no significant differences between TT and CT groups for any of the subject characteristics. All the athletes undertook twice weekly supervised lower limb strength and power training. Training programs for TT and CT groups can be observed from Tables 2 and 3, respectively. All the athletes performed 2 strength and power exercises using the squat and clean movement patterns plus additional supplementary exercises primarily focused on the abdominals, back extensors, gluteals, and hamstrings.

Only the 2 compound strength and power lifts were clustered for those in the CT group. All of these movements were executed with the intent to accelerate the load as quickly as possible for both the training groups. A mixed methods paradigm was used for the squat movement (22). This involved the use of loads ranging from 80 to 95% of 1RM for the first 6 weeks of training and a combination of heavy load squats (80–85% 1RM) and light to moderate load ballistic jump squats

TABLE 1. Subject characteristics (mean ± SD) for traditional and cluster loading groups.

	Traditional	Cluster loading
Age (y)	25.7 ± 4.5	27.8 ± 4.5
Height (m)	1.93 ± 0.1	1.85 ± 0.1
Pretraining body weight (kg)	107.3 ± 6.7	99.7 ± 10.5
Posttraining body weight (kg)	108.4 ± 6.3	100.1 ± 10.7

TABLE 2. Training program for traditional training group.*

Week	Core lifts	Set 1			Set 2			Set 3			Set 4			Set 5		
		Reps	Load (%1RM)	Rest (s)	Reps	Load (%1RM)	Rest (s)	Reps	Load (%1RM)	Rest (s)	Reps	Load (%1RM)	Rest (s)	Reps	Load (%1RM)	Rest (s)
1 and 2	Front squat	8	80	180	8	80	180	6	85	180	6	85	180	4	90	180
	Clean pull	8	80	180	8	80	180	6	85	180	6	85	180	4	90	180
3 and 4	Back squat	7	80	180	5	85	180	5	85	180	5	90	180	3	95	180
	Clean pull	7	80	180	5	85	180	5	85	180	5	90	180	3	95	180
5 and 6	Box squat	6	80	180	5	85	180	4	90	180	3	90	180	3	95	180
	Power clean	6	80	180	5	85	180	4	90	180	3	90	180	3	95	180
7 and 8	Back squat	5	80	180	4	80	180	3	85	180						
	Jump squat	4	20	180	4	10	180	3	0	180						
	Power clean	5	80	180	4	85	180	4	90	180	4	90	180	3	95	180

*Reps = repetitions; RM = repetition maximum.

(0–20% 1RM) in weeks 7 and 8. Jump squat loads were structured using a descending system with the heaviest load performed in the first set and the lightest in the final set (4). The clean pull and power clean movements used high loads (80–95% 1RM) throughout the training program. However, because the movement changed from a clean pull to a power clean for weeks 5–8 of training, the absolute load lifted in this movement pattern generally dropped considerably during the second half of the training program. There were no significant differences in the prescribed average volume load (sets $\times$ repetitions $\times$ load) per session between training groups for the squat (TT = 4.5 sets $\times$ 5 repetitions $\times$ 84.7% 1RM, CT = 4.5 sets $\times$ 4.9 repetitions $\times$ 84.7% 1RM), clean (TT = 5 sets $\times$ 4.9 repetitions $\times$ 86.5% 1RM, CT = 5 sets $\times$ 4.8 repetitions $\times$ 86.5% 1RM), or jump squat movement (3 sets $\times$ 3.7 repetitions $\times$ 10% 1RM for both groups). A total of 16 lower limb sessions were scheduled for each subject over the course of the study. An average of 99% of training was completed by the TT group, and an average of 98% of training sessions was completed by the CT group. All the sessions were supervised by a strength and conditioning coach, who stipulated training load and recorded repetitions and load completed and timed rest periods.

All the participants continued with upper body strength training (2 $\times$ per week), aerobic and anaerobic conditioning (2 $\times$ per week), speed training (2 $\times$ per week), skills training (2 $\times$ per week) and team organization (2 $\times$ per week) as part of their preseason preparation program. Average weekly training time over the course of the study was 8.5 hours. The total training load including all the components of training (resistance training, speed training, metabolic conditioning, skills training, and team organization) was periodized during the course of the study, and training load was quantified using the session rating of perceived exertion method (17). The subjects were asked to keep nutritional intake consistent through the course of the study and did not undertake supplementation additional to prescribed recovery protocols. Hydration status was assessed intermittently through the study to provide feedback to athletes on hydration status.

Statistical Analyses

All statistical analyses for force, velocity, and power variables were performed on the mean of trials 2 and 3 with the first trial excluded from analysis (25). Means and SDs were used as measures of centrality and spread of data. In the first instance, the CT and TT groups were compared using a repeated-measures 2-way ANOVA for all strength and jump squat measures. Post hoc analysis was performed using the Holm-Sidak method. An alpha level of 0.05 was used for all statistical comparisons. Additionally, the difference between the TT and CT groups was calculated (% change), and uncertainty in the effect was expressed as 90% CLs with a qualitative inference of the effect of the cluster intervention (6,7). If the confidence interval overlapped the thresholds for small positive and negative effects,

TABLE 3. Training program for cluster training group.*†

Week	Core lifts	Set 1			Set 2			Set 3			Set 4			Set 5		
		Clusters × reps	Load (%1RM)	Rest (clusters/ sets)	Clusters × reps	Load (%1RM)	Rest (clusters/ sets)	Clusters × reps	Load (%1RM)	Rest (clusters/ sets)	Clusters × reps	Load (%1RM)	Rest (clusters/ sets)	Clusters × reps	Load (%1RM)	Rest (clusters/ sets)
1 and 2	Front squat	1 × 6	80	0/180	2 × 3	80	30/120	2 × 3	85	30/120	2 × 3	85	30/120	2 × 3	90	30/120
	Clean pull	1 × 6	80	0/180	2 × 3	80	30/120	2 × 3	85	30/120	2 × 3	85	30/120	2 × 3	90	30/120
3 and 4	Back squat	1 × 5	80	0/180	1 × 5	85	0/180	1 × 2, 1 × 3	85	20–30/ 120	1 × 2, 1 × 3	90	20–30/ 120	1 × 2, 1 × 3	95	20–30/ 120
	Clean pull	1 × 5	80	0/180	1 × 5	85	0/180	1 × 2, 1 × 3	85	20–30/ 120	1 × 2, 1 × 3	90	20–30/ 120		95	20–30/ 120
5 and 6	Box squat	1 × 6	80	0/180	1 × 6	85	0/180	3 × 1	90	10/120	3 × 1	90	10/120	3 × 1	95	10/120
	Power clean	1 × 6	80	0/180	1 × 6	85	0/180	3 × 1	90	10/120	3 × 1	90	10/120	3 × 1	95	10/120
7 and 8	Back squat	1 × 5	80	0/180	1 × 4	80	0/180	1 × 3	85	0/180						
	Jump squat	2 × 2	20	20/120	2 × 2	10	20/120	3 × 1	0	10/120						
	Power clean	1 × 5	80	0/180	1 × 4	85	0/180	2 × 2	90	20/120	2 × 2	90	20/120	3 × 1	95	20/120

*Reps = repetitions; RM = repetition maximum.

†Repetitions are expressed as number of clusters × number of repetitions in each cluster, rest is expressed in seconds with the first number denoting rest between clusters and the second number rest between sets.

TABLE 4. Mean ($\pm$ SD) maximum strength, peak power, peak velocity, and peak force for the traditional and cluster groups pre and post the 8-week training intervention.

Load	Traditional		Cluster	
	Pretraining	Posttraining	Pretraining	Posttraining
Maximum strength				
Back squat 1RM (kg)	203.2 $\pm$ 16.6	240.1 $\pm$ 25.0 ^{†‡}	191.1 $\pm$ 25.0	216.4 $\pm$ 25.3 [†]
Peak power (W)				
0 kg	4,696.6 $\pm$ 460.6	4,789.6 $\pm$ 433.8	4,541.8 $\pm$ 598.9	4,716.1 $\pm$ 447.7
20 kg	4,325.5 $\pm$ 531.6	4,531.1 $\pm$ 432.3	4,143.2 $\pm$ 475.1	4,261.7 $\pm$ 305.7
40 kg	4,147.1 $\pm$ 539.6	4,168.7 $\pm$ 411.5	3,866.7 $\pm$ 306.2	4,145.5 $\pm$ 297.5
60 kg	3,942.6 $\pm$ 604.3	4,048.8 $\pm$ 504.6	3,660.3 $\pm$ 340.6	3,821.5 $\pm$ 213.2
Peak velocity (m·s ⁻¹)				
0 kg	2.18 $\pm$ 0.16	2.19 $\pm$ 0.15	2.22 $\pm$ 0.20	2.30 $\pm$ 0.17
20 kg	1.88 $\pm$ 0.16	1.92 $\pm$ 0.15	1.91 $\pm$ 0.15	1.95 $\pm$ 0.13
40 kg	1.65 $\pm$ 0.16	1.65 $\pm$ 0.15	1.65 $\pm$ 0.12	1.72 $\pm$ 0.11
60 kg	1.46 $\pm$ 0.17	1.48 $\pm$ 0.16	1.44 $\pm$ 0.09	1.49 $\pm$ 0.08
Peak force (N)				
0 kg	2,358.6 $\pm$ 140.2	2,410.6 $\pm$ 144.1	2,280.0 $\pm$ 279.8	2,329.0 $\pm$ 221.9
20 kg	2,518.8 $\pm$ 185.1	2,553.4 $\pm$ 149.3	2,393.5 $\pm$ 243.5	2,411.7 $\pm$ 208.4
40 kg	2,677.5 $\pm$ 189.6	2,671.7 $\pm$ 124.3	2,535.7 $\pm$ 199.0	2,579.2 $\pm$ 192.1
60 kg	2,837.8 $\pm$ 198.9	2,880.9 $\pm$ 155.1 [‡]	2,703.0 $\pm$ 189.7	2,708.1 $\pm$ 150.0

RM = repetition maximum.

[†]Significant within group difference pretraining to posttraining.

[‡]Significant difference between traditional and cluster posttraining.

the outcome was deemed unclear. This statistical approach has been previously used to make magnitude-based inferences in similar studies and in similar populations (1,23,36). Effect sizes (ES = pretest minus posttest divided

by the *SD* of the pretest) were also calculated for force, velocity, power, and maximum strength. Thresholds outlined by Rhea (34) specifically for highly trained athletes were used to describe effects as trivial (ES < 0.25), small (ES = 0.25–0.5), moderate (ES = 0.5–1.0), and large (ES > 1.0).

RESULTS

Mean pretraining and posttraining scores for back squat 1RM, PP, PV, and PF for both training groups can be observed from Table 4. There were significant ($p < 0.05$) increases in back squat 1RM for both the CT and TT groups (Figure 1). This increase was significantly greater ($p < 0.05$) in the TT group (% change = 18.3 ± 10.1 , ES = 2.2) compared with that in the CT group (% change = 14.6 ± 18.0 , ES = 1.0). However, the training effect of both training interventions on maximum strength was large (ES = 1.0–2.2). The PF at the external

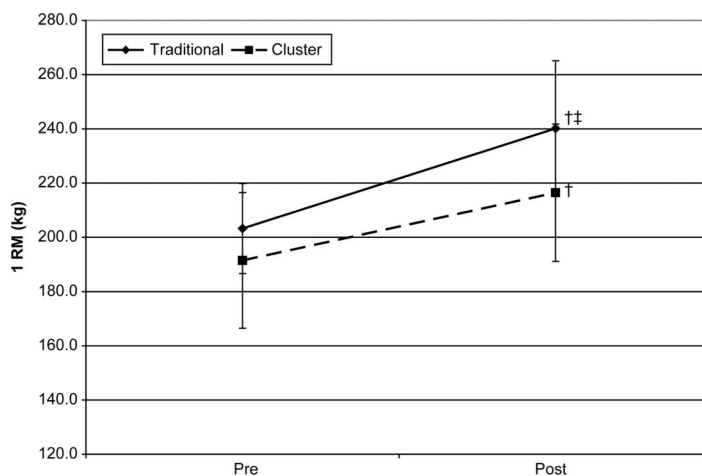


Figure 1. Back squat 1 repetition maximum (1RM) pretraining to posttraining for traditional training and cluster training groups. [†]Significant within group difference pretraining to posttraining. [‡]Significant difference between traditional and cluster posttraining.

TABLE 5. Mean percent change ($\pm SD$) and effect sizes for changes in maximum strength, peak power, peak velocity, and peak force for the traditional and cluster groups with percent difference $\pm 90\%$ CLs and qualitative practical inference of effect of the cluster intervention.

Load	Traditional		Cluster		Difference cluster – traditional	
	Mean % change $\pm SD$	Effect size	Mean % change $\pm SD$	Effect size	% Change $\pm 90\%$ CL	Qualitative inference
Maximum strength						
Back squat 1RM	18.3 $\pm$ 10.1	2.2	14.6 $\pm$ 18.0	1.0	–3.7 $\pm$ 10.5	Possibly negative
Peak power						
0 kg	2.3 $\pm$ 7.6	0.2	4.4 $\pm$ 7.1	0.3	2.2 $\pm$ 6.1	Unclear
20 kg	5.3 $\pm$ 7.4	0.4	3.5 $\pm$ 7.5	0.2	–1.8 $\pm$ 6.2	Possibly negative
40 kg	1.0 $\pm$ 6.2	0.0	7.5 $\pm$ 7.0	0.9	6.5 $\pm$ 5.6	Likely positive
60 kg	3.2 $\pm$ 5.3	0.2	4.8 $\pm$ 5.6	0.5	1.6 $\pm$ 4.5	Unclear
Peak velocity						
0 kg	0.5 $\pm$ 3.8	0.10	3.8 $\pm$ 3.4	0.40	3.3 $\pm$ 3.0	Likely positive
20 kg	2.5 $\pm$ 4.9	0.30	2.4 $\pm$ 3.9	0.30	–0.0 $\pm$ 4.4	Unclear
40 kg	0.0 $\pm$ 5.0	0.00	4.7 $\pm$ 6.1	0.60	4.7 $\pm$ 4.7	Likely positive
60 kg	1.4 $\pm$ 3.6	0.10	3.5 $\pm$ 4.7	0.50	2.1 $\pm$ 3.4	Unclear
Peak force						
0 kg	2.4 $\pm$ 6.4	0.4	2.7 $\pm$ 7.8	0.2	0.4 $\pm$ 5.9	Unclear
20 kg	1.6 $\pm$ 4.7	0.2	1.0 $\pm$ 3.4	0.1	–0.61 $\pm$ 3.4	Possibly negative
40 kg	–0.1 $\pm$ 3.1	0.0	1.8 $\pm$ 2.4	0.2	1.8 $\pm$ 2.3	Possibly positive
60 kg	1.6 $\pm$ 2.7	0.2	0.3 $\pm$ 2.8	0.0	–1.3 $\pm$ 2.6	Possibly negative

RM = repetition maximum; CL = confidence limit.

load of 60 kg was also significantly greater ($p < 0.05$) in the TT group posttraining.

Percent changes in strength and jump squat PP, PV, and PF at all loads pretraining to posttraining for TT and CT with percent difference ($\pm 90\%$ CL) between groups and a qualitative inference of the magnitude of the difference are detailed in Table 5. Percent differences between groups can be considered as clear for back squat 1RM, PP at 20 and 40 kg, PV at 0 and 40 kg, and PF at 20, 40, and 60 kg. Cluster loading had a likely positive effect for PP at 40 kg (% difference between groups = 6.5) and for PV at 0 and 40 kg (% difference between groups = 3.3 and 4.7, respectively). Additionally, the effect of the cluster intervention on PF at 40 kg was possibly positive (% difference between groups = 1.8%). The effect of the cluster intervention was possibly negative for back squat 1RM (% difference between groups = –3.7%), PF at 20 kg (% difference between groups = –0.61%), PF at 60 kg (% difference between groups = –1.3%), and PP at 20 kg (% difference between groups = –1.8%).

DISCUSSION

The purpose of this study was to ascertain whether cluster set structures provided an enhanced training stimulus for lower body strength and power development when compared with a traditional training structure during the preseason preparation of elite level rugby union players. Despite the assertion, based on acute studies focusing on cluster loading, that this

technique may be ideal for the development of mechanical power, there were no previously published studies investigating lower body power development using this training approach. We found that a traditional training structure led to greater increases in maximum strength compared with cluster structures, but that cluster training may be beneficial for improving jump squat power and velocity.

Back squat 1RM increased significantly ($p < 0.05$) in both training groups over the course of the study. However, the increases in the TT group (% change = 18.3 ± 10.1 , ES = 2.2) were significantly greater than those for the CT group (% change = 14.6 ± 18.0 , ES = 1.0). This resulted in a possibly negative effect of cluster loading on maximum strength development. Although the TT group had a small amount of extra volume prescribed, this was unlikely to be enough to significantly effect training outcomes, so it seems that set structure was most likely to be the reason for these between-group differences in adaptation. These findings are similar to those comparing traditional and cluster structures in developing upper body strength. Lawton et al. (29) reported that bench press maximum strength was increased 9.7% using a traditional set structure compared with 4.9% using a cluster structure. Our findings regarding back squat 1RM are similar to those of Lawton et al. in that maximum strength increased by 3.7% more in the traditional group. Therefore, although cluster loading was still able to elicit a large training effect for maximum strength in a highly

trained group, it seems that a traditional training structure is more effective for developing maximum strength.

The theoretical basis of cluster set structures lies in the short rest periods between clusters of repetitions allowing for metabolic recovery through the replenishment of muscular phosphocreatine, improving the quality of each effort and subsequent training adaptation (19,29,30). Although this metabolic recovery may be beneficial for quality of movement and subsequent power adaptation, it seems that the strength adaptation may benefit from the build-up of metabolites. The literature suggests significant metabolite accumulation during high load strength training protocols (13). The importance of this metabolite accumulation to adaptation is unclear (16,35); however, there is some evidence that metabolic fatigue is an important precursor to both endocrine (28,38) and neural (37,38) responses to training. Therefore, it is possible that decreased metabolite build-up during cluster loading because of the recovery between clusters is counterproductive to strength development leading to improved strength adaptation from a traditional training structure. This contention needs to be investigated using methodologies that account for the influence of metabolite accumulation on cluster loading for strength and power adaptation.

Neither training group significantly improved jump squat force, power, or velocity through the course of the study. For the TT group, all the ESs for these variables were either trivial or small. The only moderate ESs were for PP at 40 kg (ES = 0.9, % change = 7.5) and PV at 40 kg (ES = 0.6, % change = 4.7) for the CT group. Given the highly trained population that participated in the study and the relatively short training duration, this is not overly surprising. The high additional training load undertaken by the subjects during the training intervention may also have affected jump squat adaptation. This is a challenge inherent in the development of strength and power in sports such as rugby union; indeed, in a similar study and population Harris et al. (23) reported decreases in jump squat power (% change = -6 to -17.1%) and velocity (% change = -2.4 to -7.5%) despite increases in maximum strength after high and moderate load jump squat training. It has been suggested that power development may be sensitive to interference during concurrent training (27), particularly in highly trained populations. Finally, the design of the training intervention in terms of load selection may have also affected power adaptation. As is typical of resistance training prescription in collision sports, the first 6 weeks of training was focused on high load lifting. High velocity jump squats were integrated quite late in the intervention (weeks 7 and 8), which may not have allowed sufficient time for high velocity adaptation. An earlier introduction of high velocity jump squats in the training intervention may have improved the velocity and power adaptation.

We found no statistically significant difference in changes in jump squat measures pretraining to posttraining between the TT and CT groups. However, we also used CLs and magnitude-based inferences to assess the practical differences

in training outcomes between groups. With this statistical procedure, inferences were made about the true value of the effect (of cluster loading) if a large population were sampled using 90% CLs (6,7). This analysis suggested some practically positive effects in the use of cluster loading to develop power and velocity in the jump squat movement. There was a likely positive effect of CT when compared with that of TT for PP and PV at 40 kg, and for PV at body weight (Table 5). The only PP or PV value to have a greater training effect in the TT group compared with that in the CT group was PP at 20 kg (possibly negative effect for CT). Therefore, there was some evidence to support the contention, based on acute research and the importance of neural adaptation to ballistic performance, that cluster loading may be well suited to the development of velocity and power in ballistic movements. It may be that had the training intervention used in this study involved a longer ballistic (jump squat) training phase, the advantages of cluster loading for ballistic velocity and power would have been further accentuated.

The positive effect of cluster training that was apparent for jump squat PP and PV was not evident with PF. Although at 40 kg there was possibly a positive training effect for the CT group, at 20 and 60 kg, there was a possibly negative effect for CT. Previous research has also suggested that increases in moderate load jump squat PF are associated with increases in back squat maximum strength (12,32). That is, training interventions, which have a positive training outcome in terms of maximum strength development, may also increase PF in a ballistic movement such as the jump squat. Therefore, the traditional intervention, because of inducing greater maximum strength adaptation may be preferable in terms of training PF. It could therefore be concluded that to optimize ballistic power development, a combination of training methods would be optimal, a traditional intervention for development of force capabilities and cluster training for the development of velocity of movement.

PRACTICAL APPLICATIONS

Because of the importance of strength and power in collision sports such as rugby union, resistance training is an important aspect of training for athletes competing at the elite level in these sports. For these athletes, appropriate resistance training prescription is crucial for athletic development. In elite level rugby union players, cluster training structures do not provide a superior stimulus in the development of lower body maximum strength compared with a traditional loading structure. Although both a traditional structure and cluster structures could be prescribed for maximum strength, a traditional structure is likely to provide superior training outcomes. Cluster training does however present a viable training option for the development of lower body power at light to moderate external loads. Therefore, the practitioner should select the training structure, which is best suited to the individual training goals of the athlete. If high load performance and maximum strength is the key training

objective, then a traditional set structure should be used. If the development of explosive power and velocity at light to moderate loads is regarded as a more important training goal, then a cluster structure may be preferable. It may be that an integrated approach that uses both loading schemes offers optimal training adaptation or at the very least offers athlete's variation that can address training monotony.

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